AMJAD YAGHI

amjadclnyaghi@gmail.com | linkedin.com/in/amjadyaghi/ | github.com/amjadya

EDUCATION

University of British Columbia — Vancouver
Bachelor of Applied Science, Engineering Physics (Co-Op) — Expected May 2027

EXPERIENCE

Algorithms, Optimization and Control Lab, UBC — Vancouver
Research Assistant — Oct. 2026 – Present

- Building a modular timber-frame test site for an **autonomous indoor construction drone**, and characterising how its **RealSense** depth camera maps thin 2x4 framing so the planner and Gazebo sim can be tuned to fly through it reliably

UBC SAE Aero Design ↗ — Vancouver
Payload Subteam, Sensing and Docking — May 2026 – Present

- Own homing for a rover that must autonomously dock under a landed aircraft: modelled its real detectors, delays and wheels in simulation, wrote every candidate controller in **C**, and flashed the winner to the **ESP32-S3**
- Built a **constant-current LED driver** that keeps three pulsing IR beacons consistently bright as the supply sags, simulated its switching in ngspice, and wrote practice cards that teach new members the circuit

Generative Models.ai — Toronto
Software Engineer Intern — May 2025 – Aug. 2025

- Designed a **parallelized LLM blog-generation pipeline** that split outlines into per-heading prompts, enforced strict JSON schemas, and recombined validated sections into long-form drafts
- Implemented Django APIs and Temporal workflows to queue long-running analyses and **prevent request collisions**

PROJECTS

Autonomous Badminton-Playing Robot (Capstone) ↗ | *C++20, OpenCV, Eigen, CMake*

- Designed a modular **C++20 static library** (pimpl, RAII, Eigen) powering a real-time stereo vision pipeline at $\sim$**100 FPS** with hardware-synchronized cameras over GPIO triggers
- Triangulated shuttlecock 3D position across 13 m via stereo rectification and OpenCV's `triangulatePoints`, with an offline **ChArUco + court-transform calibration** workflow to map camera coordinates onto the court
- Implemented an **Extended Kalman Filter** with hand-derived analytic Jacobian for nonlinear aerodynamic drag, plus RK4 forward simulation to predict landing point within $\pm$**5 cm**
- Built a replayable set of **hand-labelled stereo flights** that scores every detector and tracker change on the same data, letting us iterate on tracking algorithms and verify the best, which caught **72%** of in-flight frames at 35 ft, up from **19%**

RL Robot-Arm Trajectory Tracking ↗ | *Python, MuJoCo, Gymnasium, Stable-Baselines3 (PPO)*

- Trained a **PPO policy** to track moving 3D targets with a simulated Franka Panda arm in **MuJoCo**; the policy outputs Cartesian lead targets, converted to joint torques via damped least-squares IK + PD control
- Beat the classical IK+PD baseline on a **200-seed paired evaluation** (mean error $-$**41%**, time-on-target **24%** $\rightarrow$ **73%**, lag **113** $\rightarrow$ **17 ms**), iterating v1 $\rightarrow$ v9 in a public build log

Autonomous Mario Kart Robot ↗ | *Arduino, STM32 (ARM), C++*

- Built a fully autonomous Arduino+STM32 Mario Kart robot using an FSM, **10 kHz** IR-beacon homing (via **convolution filtering**), and sonar edge detection for recovery; built a **zipline claw** that latched reliably to shortcut the course

SKILLS

Languages & Tools: C, C++, Python, MATLAB, SQL, 8051 assembly, Linux, Docker, git, GitHub Actions, CMake
Robotics: ROS 2, OpenCV, Eigen, Intel RealSense, FLIR Blackfly S, Gazebo, MuJoCo
Embedded: STM32, ESP32, Arduino, interrupts, ADC/DSP
Electronics: soldering, crimping, wiring harnesses, oscilloscope, digital multimeter, bench supply, Analog Discovery 2